PRACTICAL 1: Conducting a Comprehensive Literature review for a research project.

TITLE: Facial Recognition Attendance System

Shriwastav & Jain (2016) they proposed a continuous face recognition approach for student attendance, which relied on inspecting multiple image frames during class to minimize misrecognition. Implemented in MATLAB, the model compared facial features from several intervals to increase precision. They showed that this continuous inspection improved presence accuracy versus single-frame detection. The system captured multiple inputs, validated identities, and cross-checked against attendance logs. It tackled issues like motion blur or sudden lighting changes. Evaluation proved it outperformed manual verification in consistency. Their approach reinforced the utility of persistent scanning in academic spaces.

Arya et al. (2020) the authors proposed a CNN-based face recognition attendance system to replace traditional methods like Eigenfaces and Fisherfaces, which had shown limitations under poor lighting or varied postures. The system captured real-time images using a camera, processed them via CNN, and recorded attendance in Excel and MongoDB. Their method minimized human interference and enhanced data reliability. Implementation on student groups demonstrated high recognition accuracy. They emphasized real-time performance, automation, and improved system resilience. The system supported faster attendance without roll-call delays. It had been tested under varying conditions and proved consistent.

Anwar & Raychowdhury (2020) they addressed the face recognition challenges posed by masks during the COVID-19 pandemic. Their model, “MaskTheFace,” modified existing datasets by overlaying synthetic masks, then retrained FaceNet. Testing revealed a 38% improvement in true positive rates on masked datasets. The model preserved real-world face geometry and skin tone consistency. They proposed a new training pipeline for future authentication systems in health-sensitive environments. Their approach also benefited attendance systems by reducing false negatives. The research showed promising adaptability for pandemic-era identity verification.

Salac (2020) developed “PRESENT,” an Android-based attendance system using face recognition integrated with an incremental software model. The system employed a mobile camera for facial input, recognizing students through a trained model and logging attendance data locally. Faculty evaluations revealed it was “moderately acceptable” for classroom use and “highly portable” across devices. Results showed that the app reduced time spent on manual attendance. It supported classroom settings with dynamic lighting and movement. The approach favored mobile platforms due to accessibility and low cost. The project highlighted usability in live academic environments.

Irbaz et al. (2021) the authors developed a remote attendance model for employees using FaceNet and KNN classifiers. The system was trained on the LFW dataset and achieved an accuracy of 97.8%. It could be accessed via browser and used webcams to identify users in real time. The application supported remote work setups, logging attendance and timestamps without manual check-ins. Their deployment confirmed reduced error rates and high stability under different lighting and background conditions. They highlighted use cases in distributed workplaces and hybrid environments.

Mansoora et al. (2021) this team built a real-time classroom attendance system using Haar Cascades for face detection and FaceNet for recognition. Student images were captured via webcam, converted to embeddings, and matched with stored datasets. Once identified, attendance was marked and logged automatically in Excel sheets. The system reduced human intervention, speeding up attendance and improving transparency. It showed high recognition accuracy even with slight facial variations and background noise. They tested the model across different lighting conditions and camera angles. Their results confirmed strong adaptability in dynamic classrooms.

Potadar et al. (2021) colleagues proposed an AI-enhanced attendance system that integrated face recognition using a combination of machine learning and image processing techniques. The project emphasized real-time data processing and cloud syncing for attendance reports. They focused on cost-efficiency, ease of deployment, and high recognition accuracy. Results indicated that the proposed model outperformed fingerprint and manual systems in speed and reliability. The authors also discussed future use of predictive analytics and behavior tracking. The model supported multi-class integration and was scalable for large institutions. Their study marked a shift toward intelligent attendance monitoring.

Rao (2022) introduced “Attend Face,” a fully automated system capturing multiple facial images during class at fixed intervals. It checked attendance by matching faces with pre-enrolled student profiles every ten minutes. Students had to be identified in at least 60% of frames to be marked present. It supported concurrent classes and offered Moodle integration for academic records. The system operated with minimal human involvement and resisted spoofing attempts. Tests across different classroom sizes showed excellent scalability and precision. The author highlighted its potential in hybrid or fully online education.

Sharma (2022) developed a novel attendance system combining mask detection and face recognition using MobileNetV2. The framework included real-time monitoring with Open CV and a GUI-based dashboard. It first confirmed whether a user wore a mask, then identified the face to record attendance. The system addressed pandemic-induced constraints and ensured compliance with health guidelines. They observed effective performance under constrained face visibility and varying user angles. The interface supported easy report generation and export. It demonstrated accurate results in semi-controlled environments like classrooms and offices.

Prasad et al. (2023) this team surveyed various face recognition techniques like PCA, ICA, SVM, and Genetic Algorithms for classroom attendance. They reviewed literature spanning real-time image processing, model accuracy, and robustness to occlusions and lighting. Their findings suggested PCA and ICA were faster but less accurate, while SVM offered better classification at a higher computational cost. They emphasized that hybrid models showed promise in real world deployment. The paper provided insights into choosing appropriate algorithms based on environmental and institutional constraints.

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